Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years shipping consumer and internal tools in **TypeScript/React** and **Python**/Node.js, with **trust and safety** roots in phishing and malware URL detection, payment fraud and dispute flows, and AI-native development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Translated data-access requirements into a multi-tenant schema on Azure SQL with **row-level security** for 500+ users, isolating each tenant's records and cutting dashboard latency from 1.2s to 350ms.
- Built fault-tolerant **data pipelines** and LLM job workers with FastAPI, validation, and reconciliation checks, processing 1K+ AI jobs at 3x throughput and 5K+ monthly integrations at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a full-stack **React** and Flask platform with AI-assisted grading and feedback using GPT-4o, Claude, and Qwen-VL, serving 150+ instructors and 10K+ daily requests while cutting turnaround 60%.
- Designed **secure file ingestion** for PDF, DOCX, and ZIP uploads with Flask-Login authentication, concurrent batch processing of 1K+ files, and per-call model usage logs for usage and cost auditability.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built internal CRM tooling with React, TypeScript, Node.js, and MongoDB, including customer profiles with interaction history, lead tracking, and **role-based access**, syncing 2M+ records in real time via WebSockets.
- Added JWT authentication, Snyk and OWASP dependency scanning, and Jest, Mocha, and Selenium tests to Jenkins **CI/CD**, reaching 95%+ build validation and triaging production errors with Sentry.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst – Research Fellowship *India*

- Built a **malicious URL detection** pipeline over 650K+ URLs with 21 behavioral indicators such as IP presence, URL length, and suspicious keywords to surface phishing and malware patterns.
- Compared Gaussian Naive Bayes, Random Forest, LSTM, Bi-LSTM, GRU, and CNN classifiers for **threat detection**, reaching 96%+ accuracy and producing actionable insights into phishing and malware trends.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Integrated PCI-compliant multi-currency payment vendor APIs with **fraud checks** and idempotent webhooks for refunds, disputes, and chargebacks, keeping payment status consistent and cutting manual refund handling 70%.
- Implemented passwordless wallet-signature **authentication** and JWT sessions, eliminating password storage for wallet users, and used Firebase transactional locking to prevent double-booked time slots across 50K+ booking records.

Skills

Languages: TypeScript, JavaScript, Python, SQL, Java
Full Stack: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets, TailwindCSS
Trust & Safety: Fraud and Dispute Workflows, Phishing/Malware Detection, Feature Engineering, Random Forest, LSTM, RBAC, JWT, Row-Level Security
AI: Claude, Claude Code, GPT-4o, LLM Pipelines, Prompt Chaining, OCR
Data & Integrations: PostgreSQL, MongoDB, Firebase, Supabase, Databricks, Webhooks, Payment APIs, Twilio
Cloud & Quality: AWS, Azure, Docker, CI/CD, Jest, Selenium, Sentry

Projects

Trust and Safety Case Console | *Next.js, TypeScript, Node.js, PostgreSQL, Claude, Python*

- Designed a case-management console in Next.js and Node.js where Claude triages reports, a **URL-risk model** flags scam links, reviewers act through RBAC-gated queues, and every decision lands in an **audit log**.

FinNexus Live | *Next.js, TypeScript, MongoDB, Plaid, WebSockets*

- Built a consumer **fintech** app in TypeScript and Next.js with Plaid bank linking, a token-based firewall protecting transaction access, and real-time WebSocket updates for 50K+ users and 1M+ transactions.

HMIS – Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Twilio, Sentry*

- Won Smart India Hackathon with a TypeScript app turning patient-consent rules into **consent-gated** record access via ABDM APIs, plus dashboards, Twilio alerts, and Sentry monitoring for 100+ daily appointments.